Rafay Farah

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Education

University of Washington Expected June 2028
Bachelor of Science in Computer Science *Bothell, WA*

- **GPA: 3.99/4.00** | Dean's List
- **Relevant Courses:** Data Structures & Algorithms, Object-Oriented Programming, Discrete Mathematics, Database Systems, Software Engineering, Software Design, Statistics, Calculus I-III

Experience

Undergraduate Researcher Aug 2026 – Present
Carnegie Mellon University, S3D *Pittsburgh, PA*

- Build automated Python pipelines that test how accurately AI models turn hand-drawn sketches into code
- Debug evaluation workflows and investigate recurring model failures across visual programming tasks
- Design a follow-up study testing whether newer models can beat a prior study's 36% accuracy baseline
- Develop evaluation methods for newer multimodal models and a potential voice-plus-sketch input mode

AI Research Engineer Intern June 2026 – Aug 2026
University of Pittsburgh, LRDC FACET Lab *Pittsburgh, PA*

- Built and deployed Cora, a full-stack **Next.js/TypeScript** tutoring platform now being tested in several Pennsylvania high schools, implementing user-facing features, and preparing the product for classroom use
- Integrated **Claude**, **Gemini**, and **OpenAI APIs** behind a unified model workflow, debugging application behavior across tutoring and evaluation tasks and improving reliability across AI-guided interactions
- Collaborated through **Git**, pull requests, code reviews, testing, and weekly team design discussions to implement new features, resolve issues, and improve software quality before classroom deployment and field testing

Lead Software Engineer Jan 2026 – June 2026
Startup Club *Bothell, WA*

- Led the club's software team across multiple internal products, building and shipping tools for company research, outreach, and operations while guiding members through implementation and technical decisions
- Built a **React/TypeScript** company-discovery platform with database-backed search and 3+ API integrations, helping evaluate **500+** companies and reduce manual research time by **40%** while improving outreach
- Translated **10+** product requirements and user stories into scoped engineering tasks, iterating through weekly **Agile** development cycles to deliver features more efficiently and align the product with stakeholder needs

Hackathons & Projects

Stride | Cal Hacks 13.0 | *TypeScript, Electron, Swift, Claude, Playwright* | GitHub

- Built a macOS AI agent capable of performing virtually any everyday computer task, interpreting natural-language requests, on-screen instructions, and documents to autonomously execute workflows across applications
- Engineered a multi-agent computer-use system integrating **30+** tools for desktop control, browser automation, voice interaction, and troubleshooting, with built-in safety enforcement and real-time result verification

HeadsUp | HackGT 13 | *Next.js, TypeScript, Python, DuckDB, Supabase, CMS/RxNorm* | GitHub

- Built a Medicare drug-coverage monitoring platform for Impiricus's HackGT challenge that compares CMS Part D releases and alerts physicians when coverage changes affect medications they prescribe
- Built **SQL** pipelines to detect coverage changes across **111K+** formulary records, **152** Georgia plans, and **1M+** pricing records; integrated **Supabase** authentication and doctor-specific data access

Insta Insights | HackMIT 2026 | *Next.js, TypeScript, Elasticsearch, Python, Meta Muse Spark* | GitHub

- Built a full-stack app that parses and normalizes a user's own Instagram data export (41K+ messages, 400+ chats) to surface fading friendships, unanswered messages, and unfinished plans worth following up on
- Engineered an **Elasticsearch** ranking pipeline with deterministic scoring, closeness modeling, and **BM25** fallback; integrated **Meta Muse Spark** with source-message validation and typed **Next.js** APIs

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Spring Boot, PyTorch, NumPy
Platforms & Tools: Git, GitHub, Linux, Docker, AWS, REST APIs, PostgreSQL, Supabase, Claude Code
Interests: Badminton, Weightlifting, Gaming, Traveling, Content Creation, Hackathons